2)] $x_0: Q_m(x_0)=0 \Rightarrow Q_m(\bar{x}_0)=0$.] $x_0=\alpha+i\beta$ $(\beta\neq 0)$

Э $(x-x_0)(x-\bar{x}_0)=(x-\alpha-i\beta)(x-\alpha+i\beta)=$

$=(x-\alpha)^2+\beta^2=x^2-2\alpha x+(\alpha^2+\beta^2)=$

$=\underline{x^2+px+q}$, $p=-2\alpha$, $q=\alpha^2+\beta^2$

$p^2-4q=-4\beta^2<0 \Rightarrow Q_m(x)=(x^2+px+q)\tilde{Q}_{m-2}(x)$

3)] $x_0=\alpha+i\beta$ $(\beta\neq 0)$ – корень кр-ти $\underline{s}$ $\Rightarrow$

$Q_m(x)=(x-x_0)^s(x-\bar{x}_0)^s\tilde{Q}_{m-2s}(x)=$

$=(x^2+px+q)^s\tilde{Q}_{m-2s}(x)$, $\tilde{Q}_{m-2s}(x_0)\neq 0$

$\Rightarrow$ Т] $Q_m(x)$: $C_m=1$ $\Rightarrow$

$$Q_m(x)=(x-x_1)^{k_1}(x-x_2)^{k_2}\dots(x-x_p)^{k_p}\cdot$$
$$\cdot(x^2+p_1x+q_1)^{l_1}(x^2+p_2x+q_2)^{l_2}\dots(x^2+p_sx+q_s)^{l_s}$$

где $x_i\in\mathbb{R}$, $p_j, q_j\in\mathbb{R}$, $i=\overline{1,p}$, $j=\overline{1,s}$

$k_i\in\mathbb{N}$, $l_j\in\mathbb{N}$, $p_j^2-4q_j<0$, $\sum_{i=1}^{p}k_i+2\sum_{j=1}^{s}l_j=m$

Л1] $\frac{P_n(x)}{Q_m(x)}$ - пр. рац. др.] $x=a\in\mathbb{R}$, $Q_m(a)=0$, $k\geq 1$ – кр

$\Rightarrow \exists A\in\mathbb{R}$, $\exists \tilde{P}(x)$:

$$\frac{P_n(x)}{Q_m(x)}=\frac{A}{(x-a)^k}+\frac{\tilde{P}(x)}{(x-a)^{k-1}Q^{(1)}_{m-k}(x)} \quad (*)$$

– пр. р. др, где

$Q_m(x)=(x-a)^kQ^{(1)}_{m-k}(x)$, причём $\odot$!

Δ (∃) Э $\frac{P_n(x)}{Q_m(x)}-\frac{A}{(x-a)^k}=\frac{P_n(x)}{(x-a)^kQ^{(1)}_{m-k}(x)}-\frac{A}{(x-a)^k}=$

$=\frac{P_n(x)-AQ^{(1)}_{m-k}(x)}{(x-a)^kQ^{(1)}_{m-k}(x)}$. Выберем A: $P_n(a)-AQ^{(1)}_{m-k}(a)=0$

$\Rightarrow A=\frac{P_n(a)}{\underset{\neq 0}{Q^{(1)}_{m-k}(a)}} \Rightarrow P_n(x)-AQ^{(1)}_{m-k}(x)=(x-a)\tilde{P}(x)$

$\Rightarrow (*)$.

(!)] $\exists$ $\frac{P_n(x)}{Q_m(x)}=\frac{A_1}{(x-a)^k}+\frac{\tilde{P}_1(x)}{(x-a)^{k-1}Q^{(1)}_{m-k}(x)}=\frac{A_2}{(x-a)^k}+\frac{\tilde{P}_2(x)}{(x-a)^{k-1}Q^{(1)}_{m-k}(x)}$

$\cdot(x-a)^kQ^{(1)}_{m-k}(x) \Rightarrow A_1\cdot Q^{(1)}(x)+\tilde{P}_1(x)(x-a)=A_2\cdot Q^{(1)}(x)+\tilde{P}_2(x-a)$

это верно $\forall x \in \mathbb{R}$. $\exists$ $\underline{x = a}$ $\Rightarrow$

$$A_1 \cdot \underset{\neq 0}{Q^{(1)}(a)} = A_2 \underset{\neq 0}{Q^{(1)}(a)} \Rightarrow A_1 = A_2$$

$\Rightarrow$ $P_n(x) - A\,Q^{(1)}_{m-k}(x)$ — опр. ! — но •

Т.к. $P_n(x) - A\,Q^{(1)}_{m-k}(x) = (x-a)\tilde{P}(x)$, и

$\deg \tilde{p}(x)(x-a) \leq \max(n, m-k)$
$n < m$, $m-k \leq m-1 < m$ $\Rightarrow$

$\deg \tilde{P}(x)(x-a) < m$ ▲

Следствие Применив (Л1) k-раз, получим

$$\frac{P_n(x)}{Q_m(x)} = \frac{A_k}{(x-a)^k} + \frac{A_{k-1}}{(x-a)^{k-1}} + \ldots + \frac{A_1}{x-a} + \frac{\tilde{P}(x)}{Q^{(1)}_{m-k}(x)}$$

$A_i \in \mathbb{R}$, $\tilde{P}(x)$ с $\mathbb{R}$-ми к-ми

$\frac{\tilde{P}}{Q^{(1)}_{m-k}}$ - пр., $Q^{(1)}_{m-k}(a) \neq 0$

(Л2) $\exists$ ① $\frac{P_n(x)}{Q_m(x)}$ - пр рац. др ② $Q_m(\alpha + i\beta) = 0$, кр $\underline{k \geq 1}$, т.е

$Q_m(x) = (x^2 + px + q)^k \tilde{Q}_{m-2k}(x)$, $\tilde{Q}_{m-2k}(\alpha + i\beta) \neq 0$

$p^2 - 4q < 0$ $\Rightarrow$ $\exists$ $A, B \in \mathbb{R}$, $\tilde{P}(x)$:

$$\frac{P_n(x)}{Q_m(x)} = \frac{Ax+B}{(x^2+px+q)^k} + \frac{\tilde{P}(x)}{(x^2+px+q)^{k-1}\tilde{Q}_{m-2k}(x)} \quad (**)$$

— пр.

это пр-ние !

Δ ① 9 $\frac{P_n(x)}{Q_m(x)} - \frac{Ax+B}{(x^2+px+q)^k} = \frac{P_n(x) - (Ax+B)\tilde{Q}(x)}{(x^2+px+q)^k \tilde{Q}(x)}$

A и B: $P_n(\alpha + i\beta) - (A(\alpha + i\beta) + B)\tilde{Q}(\alpha + i\beta) = 0$

$\Rightarrow$ $A\alpha + B + iA\beta = \underbrace{\frac{P_n(\alpha + i\beta)}{\tilde{Q}(\alpha + i\beta)}}_{= R}$

$\Rightarrow \begin{cases} A\alpha + B = \operatorname{Re} R \\ A\beta = \operatorname{Im} R \end{cases}$

$\Downarrow$

$$A = \frac{\operatorname{Im} R}{\beta}, \quad B = \operatorname{Re} R - \frac{\alpha \operatorname{Im} R}{\beta}$$

$$\Rightarrow P_n(x) - (Ax+B)\tilde{Q}(x) = (x^2+px+q)\tilde{P}(x) \Rightarrow (**).$$

• ! $\exists$ $$\frac{P_n(x)}{Q_m(x)} = \frac{A_1x+B_1}{(x^2+px+q)^k} + \frac{\tilde{P}_1(x)}{(x^2+px+q)^{k-1}\tilde{Q}(x)} =$$

$$= \frac{A_2x+B_2}{(x^2+px+q)^k} + \frac{\tilde{P}_2(x)}{(x^2+px+q)^{k-1}\tilde{Q}(x)}; \quad * (x^2+px+q)^k\tilde{Q}(x) \Rightarrow$$

$$(A_1x+B_1)\tilde{Q}(x) + \tilde{P}_1(x)(x^2+px+q) =$$

$\forall x$

$$= (A_2x+B_2)\tilde{Q}(x) + \tilde{P}_2(x)(x^2+px+q) \Rightarrow$$

$\exists$ $\underline{x = \alpha + i\beta}$ $\Rightarrow (A_1(\alpha+i\beta)+B_1)\tilde{Q}(\alpha+i\beta) =$ $\ne 0$

$$= (A_2(\alpha+i\beta)+B_2)\tilde{Q}(\alpha+i\beta)$$

$\ne 0$

$$\Rightarrow \begin{cases} A_1\alpha + B_1 = A_2\alpha + B_2 \\ A_1\beta = A_2\beta \end{cases} \Rightarrow \begin{cases} A_1 = A_2 \\ B_1 = B_2 \end{cases}$$

$\Rightarrow$ $P_n(x) - (Ax+B)\tilde{Q}(x)$ – опр ! – но пр – ние

$\exists$ $r = \deg[(x^2+px+q)\tilde{P}(x)]$, то

$$\begin{cases} r \le n \\ r \le m-2k+1 \end{cases} \Rightarrow r \le m-1 \Rightarrow \frac{\tilde{P}(x)}{(x^2+px+q)^{k-1}\tilde{Q}(x)} \text{ – прав}$$

$r \le \max(n, m-2k+1)$

$\blacktriangle$

<u>Следствие</u> применив $(Л_2)$ $\underline{k}$ раз, получим

$$\frac{P_n(x)}{Q_m(x)} = \frac{A_1x+B_1}{x^2+px+q} + \frac{A_2x+B_2}{(x^2+px+q)^2} + \dots + \frac{A_kx+B_k}{(x^2+px+q)^k}$$

$$+ \frac{\tilde{P}(x)}{\tilde{Q}_{m-2k}(x)}, \quad A_i, B_i \in \mathbb{R}, \ \tilde{P}(x) \text{ с } \mathbb{R} \text{ – ми } k\text{ – ми}$$

– прав $\tilde{Q}(\alpha+i\beta) \ne 0$